

Table of Contents

Problem #1

Handwritten Work _____ Pgs 2-5

Matlab code _____ Pg 6

Problem #2

Matlab Output _____ Pgs 7-8

Bode/Gain and Phase margin plots _____ Pg 10

Matlab Code _____ Pgs 11-12

Problem #1

$$* G(s) = \frac{10}{s(s+1)(s+2)} \Rightarrow \text{Zeros} = \text{N/A} \quad \text{Poles} = 0, -1, -2$$

gain = 10

$\Rightarrow$ break point frequencies @ 0, 1, 2

$$G(j\omega) = \frac{10}{2} \frac{1}{(j\omega)(\frac{j\omega}{1} + 1)(\frac{j\omega}{2} + 1)} \approx \frac{1}{(j\omega)(\frac{j\omega}{1} + 1)(\frac{j\omega}{2} + 1)}$$

$$\Rightarrow 20 \log(G(j\omega)) = 20(\log(s) - \log(\frac{j}{\omega}) - \log(\frac{j\omega}{1} + 1) - \log(\frac{j\omega}{2} + 1))$$

constant gain of $\frac{10}{2} = 5 \Rightarrow 20 \log(5) = 13.979 \text{ dB}$

$\Rightarrow$ a magnitude shift of $\pm 14 \text{ dB}$ of the simplified $G_s(j\omega) = \frac{1}{(j\omega)(\frac{j\omega}{1} + 1)(\frac{j\omega}{2} + 1)}$

$\therefore$ Magnitude response of $1/s \Rightarrow |G(j\omega)| = 20 \log(\frac{1}{\omega}) = -20 \log(\frac{1}{\omega})$

$\Rightarrow$ as the ω approaches 0 from the right the phase $\approx -90^\circ$

$\Rightarrow$ phase plot starts @ $\approx 90^\circ$ and constant ($0 < \omega < 0.1$)

and the magnitude is -20 dB/decade from break point frequency $0 < \omega$

$\therefore$ Magnitude response of $\frac{a}{s+a} \Rightarrow |G_s(j\omega)| \Big|_{a=1} = 20 \log(\frac{1}{\omega}) = -20 \log(\frac{a}{\omega})$

$\Rightarrow$ phase shifts $0 = -90^\circ/\text{decade}$ before and after $a = 1$

the magnitude is -20 dB/decade for values $\omega > a$, phase = -90°

for values $\omega < a$, phase = 0°

for $|G_s(j\omega)| \Big|_{a=2} = 20 \log(\frac{2}{\omega}) = -20 \log(\frac{2}{\omega})$

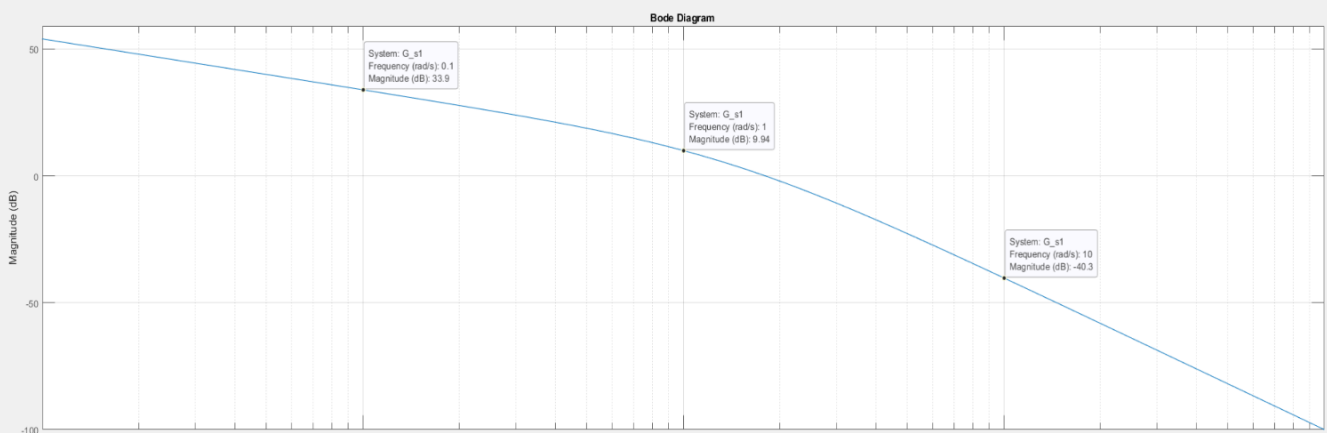
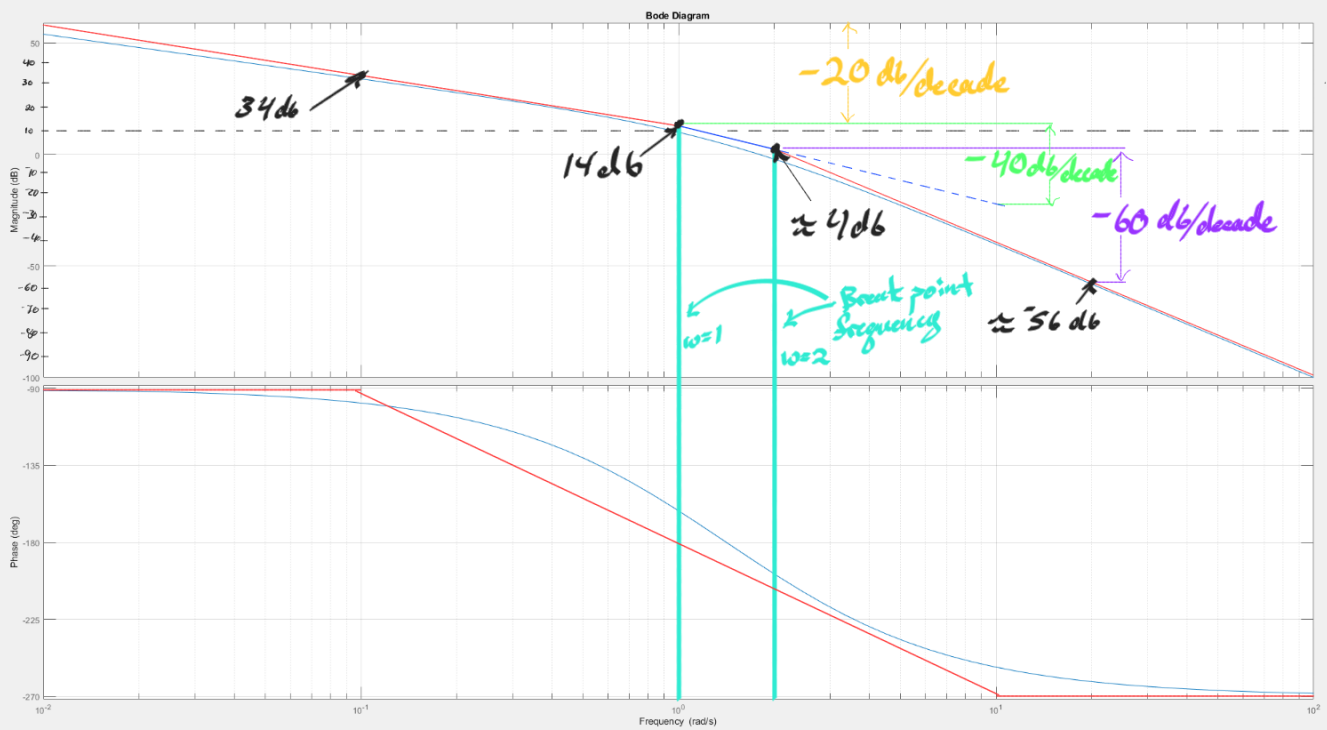
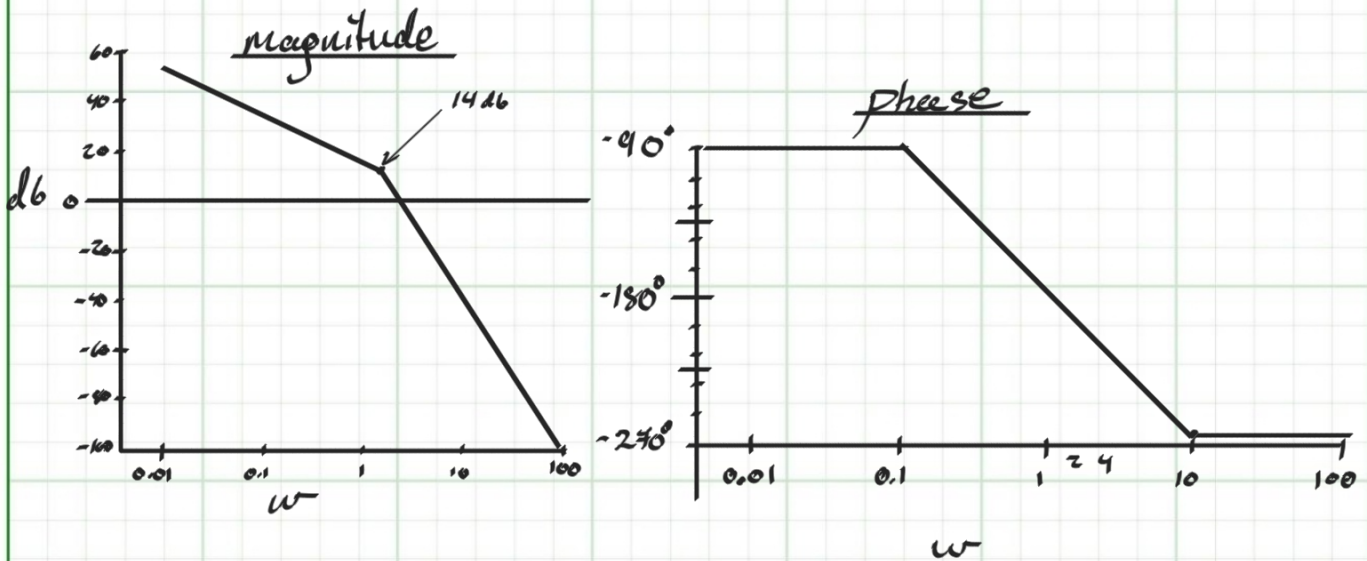
$\Rightarrow$ the magnitude is -20 dB/decade

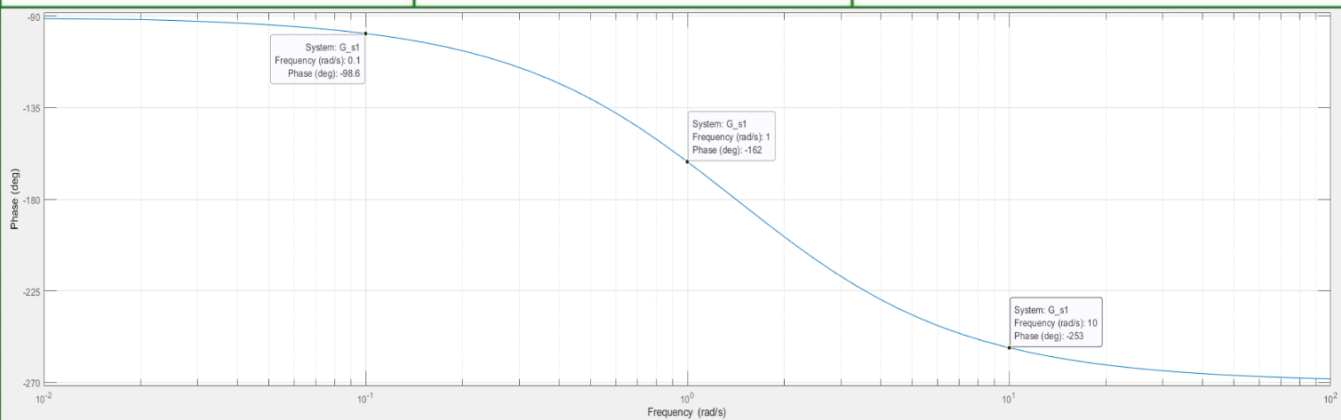
$\Rightarrow |G(j\omega)| \Rightarrow \begin{cases} \therefore 0 < \omega < 1 = -20 \text{ dB/decade} \\ \therefore 1 < \omega < 2 = -20 \text{ dB/decade} \\ \therefore 2 < \omega = -20 \text{ dB/decade} \end{cases} \Rightarrow \begin{cases} 1 < \omega < 2 = -40 \text{ dB/decade} \\ 2 < \omega = -60 \text{ dB/decade} \end{cases}$

$\therefore$

ω	0	0.1	≈ 1	2	10^2
Phase/decade	0	-90°	-90°	-90°	0
Mag/decade		-20 dB	-40 dB	-60 dB	-60 dB

Mag Shift. $\uparrow = 14\text{ dB}$





$$* G(s) = \frac{50(s+0.1)}{s(s+2)(s+4)} \Rightarrow \text{zeros} = 0.1 \quad \text{poles} = 0, -2, -4$$

gain = 50

Break point frequencies @ 0.1, 2, 4

$$G(s) \Rightarrow \frac{50(s+0.1)}{8s(\frac{s}{2}+1)(\frac{s}{4}+1)} \Rightarrow G(j\omega) = \frac{50}{8} \left(\frac{1}{j\omega} \right) \left(\frac{s+0.1}{1} \right) \left(\frac{1}{\frac{j\omega}{2}+1} \right) \left(\frac{1}{\frac{j\omega}{4}+1} \right)$$

$$\Rightarrow 20 \log(50/8) = 15.92 \text{ dB}$$

$$-20 \log\left(\frac{1}{\omega}\right) \Rightarrow \text{phase starts @ } \pm 90^\circ, \text{ mag } -20 \text{ dB}$$

$$20 \log\left(\frac{\omega}{a}\right) \Big|_{\omega=a}^{a=0.1} \Rightarrow \text{phase} = 90^\circ \text{ 1 decade before and after } \omega=0.1$$

mag = 20 dB/decade

$$-20 \log\left(\frac{a}{\omega}\right) \Big|_{\omega=2}^{a=2} \text{ phase} = -90^\circ \text{ 1 decade before and after } \omega=2$$

mag = -20 dB/decade $2 < \omega < 4$

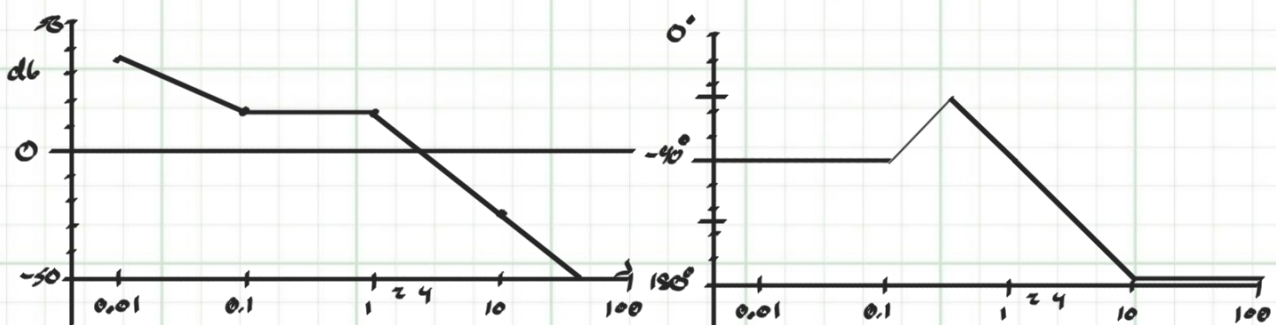
$$-20 \log\left(\frac{a}{\omega}\right) \Big|_{\omega=4}^{a=4} \Rightarrow \text{phase of } -90^\circ \text{ 1 decade before and after } \omega=10$$

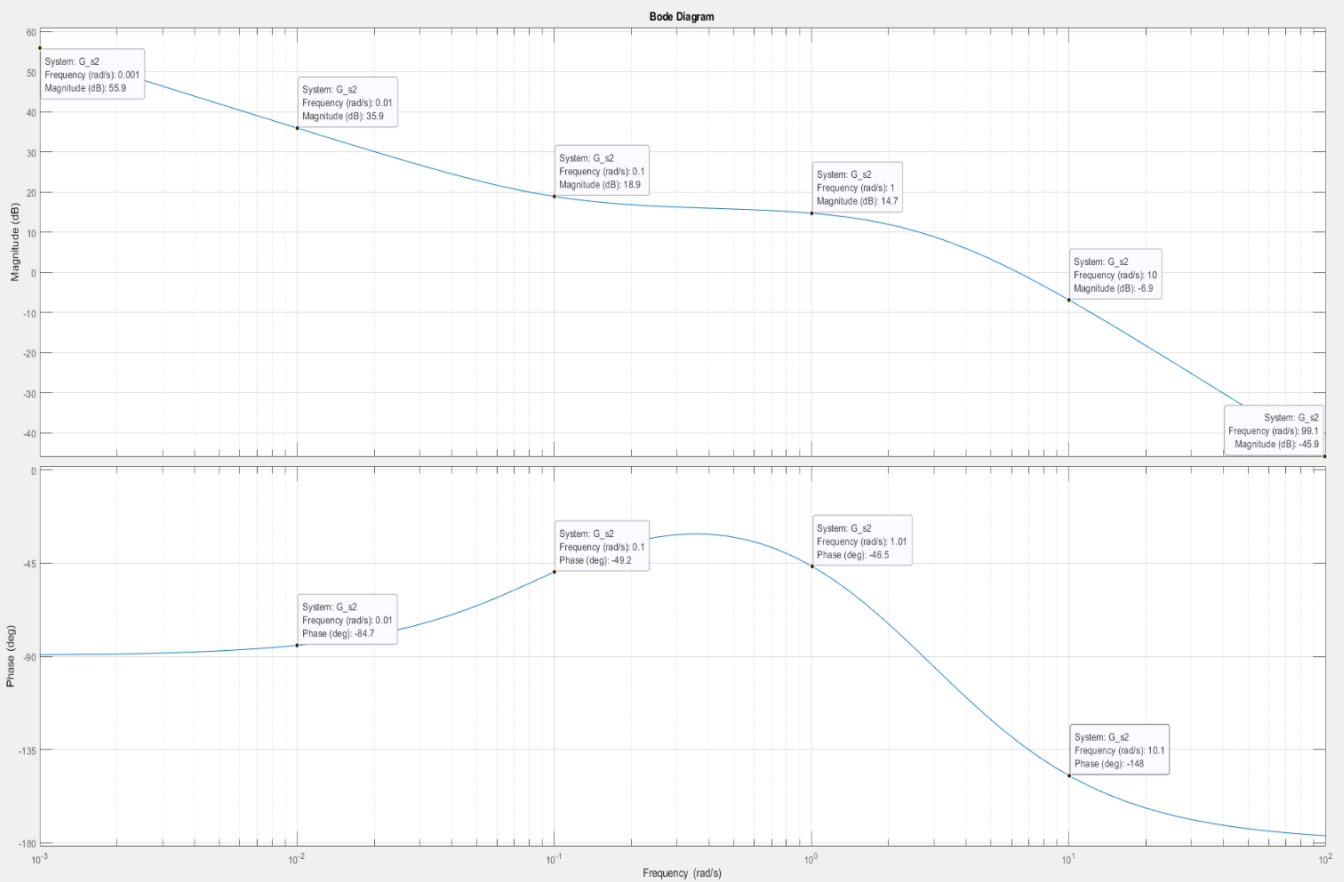
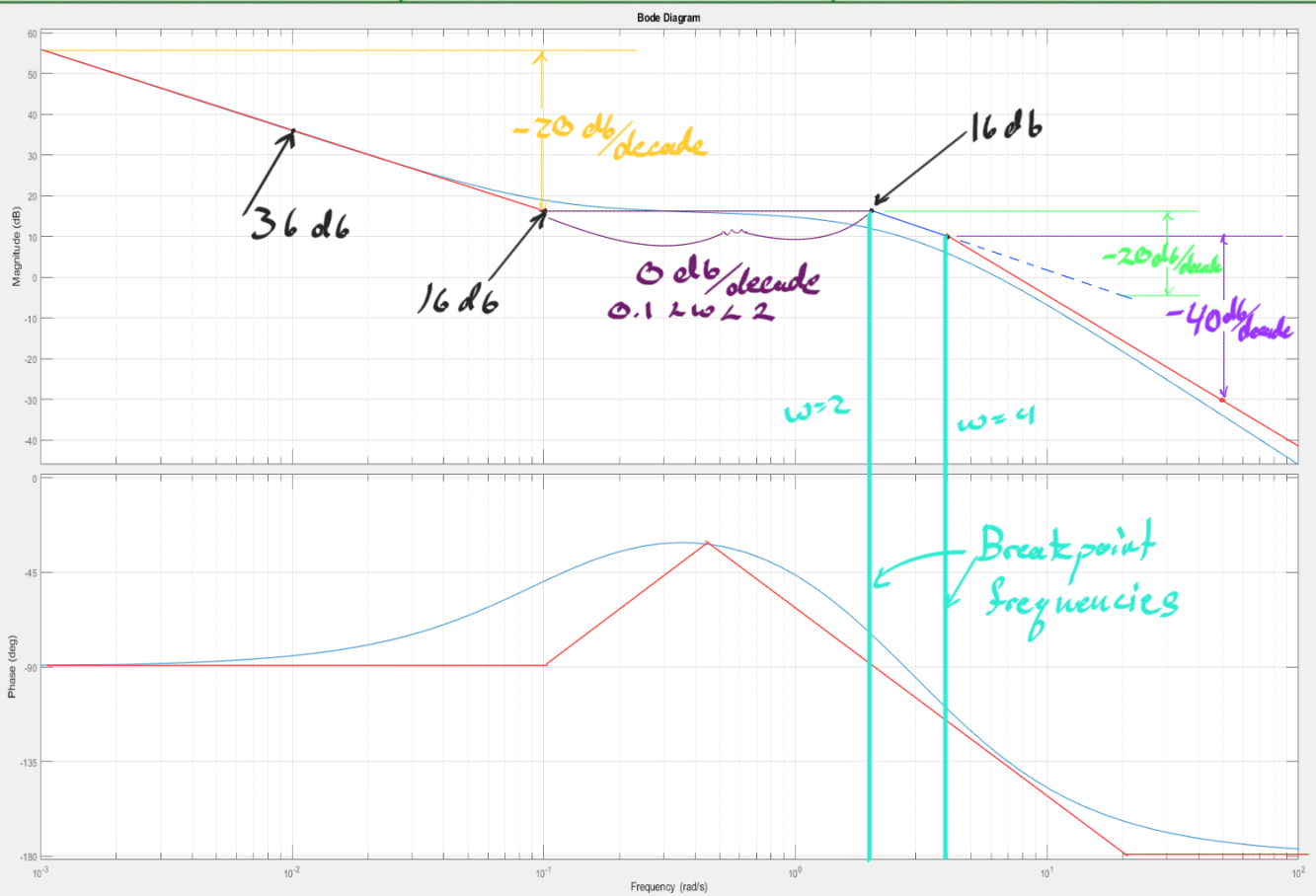
mag = -20 dB/decade

$$\Rightarrow \omega < 0.1 \quad 0.1 < \omega < 2 \quad 2 < \omega < 4 \quad 4 < \omega$$

phase $-90^\circ \quad 0^\circ \quad -90^\circ \quad -90^\circ$

mag $-20 \text{ dB/decade} \quad 0 \text{ dB/decade} \quad -20 \text{ dB/decade} \quad -40 \text{ dB/decade}$





```
clc
clf
clear
```

```
G_s1=zpk([], [0 -1 -2], 10)
G_s2=zpk([-0.1], [0 -2 -4], 50)
figure(1)
bode(G_s1)
figure(2)
bode(G_s2)
```

MATLAB Command Window

G_s1 =

$$\frac{10}{s(s+1)(s+2)}$$

Continuous-time zero/pole/gain model.
Model Properties

G_s2 =

$$\frac{50(s+0.1)}{s(s+2)(s+4)}$$

Continuous-time zero/pole/gain model.
Model Properties

>>

Considering Unity Feedback systems with open loop transfer Functions in the form of $K*G(s)$

This Program will populate a 'Bode Plot' for any number of transfers function with calculated and displayed values of the gain and phase margin based on values of input K have the following ready for input:

*Value of constant gain K

input K in the form: [K]

*Transfer Functions -- zero(s) and pole(s)

Input zeros in the form: [Z1 Z2 Z3 ... Zn] i.e. numerator is in the form of:
(s '+' or '-' Z1)(s '+' or '-' Zn)

if numerator has no zeros input []

Input poles in the form: [P1 P2 P3 ... Pn] i.e. denominator is in the form of:
(s '+' or '-' P1)(s '+' or '-' Pn)

if dinominator has no zeros input []

*If transfer fucntion is in the ploynomial form:

$K*G(s) = K(CN1*s^{(n-1)}+CN2*s^{(n-2)}+CNn*s^{(0)} / (CD1*s^{(n-1)}+CD2*s^{(n-2)}+CDn*s^{(0)})$

Where CN denotes Numerator constant value & CD denotes Denominator constant

value

Input values for CN and CD in the form [C1 C2 C3... Cn]

How many transfer functions do you want to evlauate: 2

Is transfer Function in polynomial of Y/N: Y

Input single nummmerical value for K: 40

Input Numerator constant values: [1 5]

Input Denominator constant values: [1 4 25 0]

G_s =

40 s + 200

s^3 + 4 s^2 + 25 s

Continuous-time transfer function.

Model Properties

Gs =

40 (s+5)

s (s^2 + 4s + 25)

Continuous-time zero/pole/gain model.

Model Properties

Gm =

2.5006

Pm =

8.3814

Wcg =

11.1815

Wcp =

7.7997

Is transfer Function in polynomial of Y/N: N

Input single numerical value for K: 40

Input zero values Z:

Input pole values: [0 -3 -12]

G_s =

From input 1 to output:

$40 s + 200$

 $s^3 + 4 s^2 + 25 s$

From input 2 to output:

40

 $s^3 + 15 s^2 + 36 s$

Continuous-time transfer function.

Model Properties

Gs =

40

 $s (s+12) (s+3)$

Continuous-time zero/pole/gain model.

Model Properties

Gm =

13.5000

Pm =

65.8119

Wcg =

6

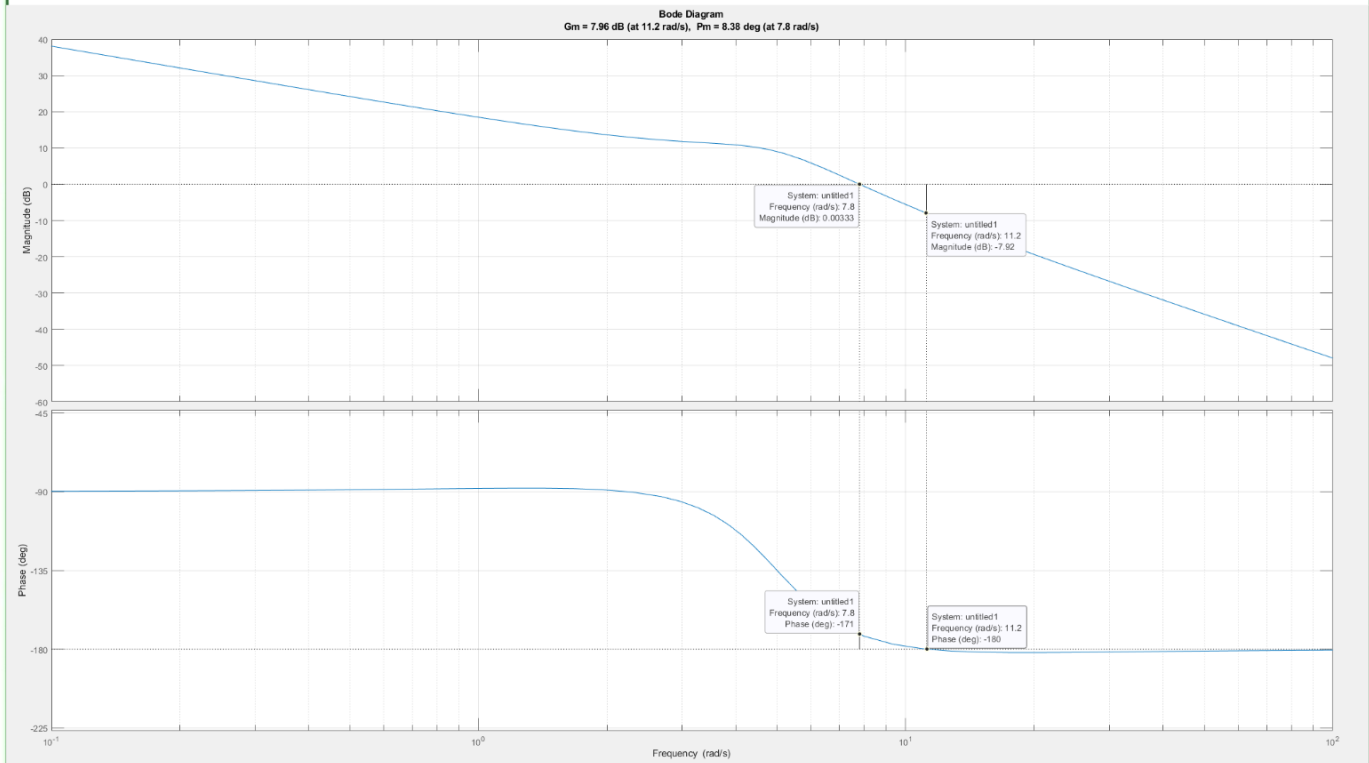
Wcp =

1.0453

>>

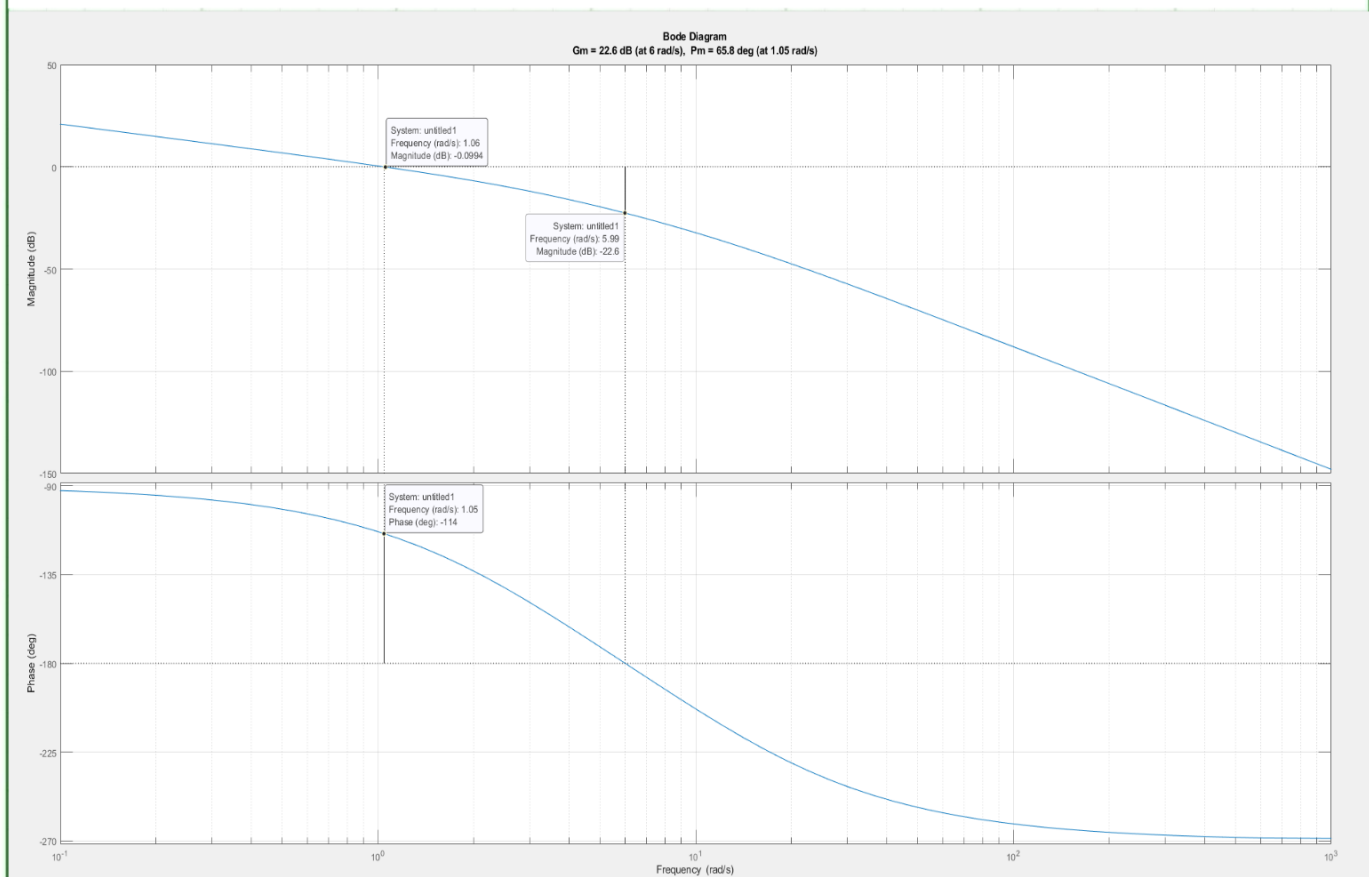
$G_s =$

$$\frac{40 (s+5)}{s (s^2 + 4s + 25)}$$



$G_s =$

$$\frac{40}{s (s+12) (s+3)}$$



```

clc
clear
clf

fprintf("Considering Unity Feedback systems with open loop transfer Functions in the form
of K*G(s)\n\n" + ...
    "This Program will populate a 'Bode Plot' for any number of transfers function
with calculated and displayed values\n" + ...
    "of the gain and phase margin based on values of input K have the following ready
for input:\n\n" + ...
    "    *Value of constant gain K\n" + ...
    "        input K in the form: [K]\n" + ...
    "    *Transfer Functions -- zero(s) and pole(s)\n" + ...
    "        Input zeros in the form: [Z1 Z2 Z3 ... Zn] i.e. numerator is in the form
of:\n" + ...
    "            (s '+' or '-' Z1) (s '+' or '-' Zn)\n\n" + ...
    "            if numerator has no zeros input []\n\n" + ...
    "        Input poles in the form: [P1 P2 P3 ... Pn] i.e. denominator is in the
form of:\n" + ...
    "            (s '+' or '-' P1) (s '+' or '-' Pn)\n" + ...
    "            if dinominator has no zeros input []\n\n" + ...
    "    *If transfer fuction is in the ploynomial form:\n" + ...
    "        K*G(s) = K(CN1*s^(n-1)+CN2*s^(n-2)+CNn*s^(0) / (CD1*s^(n-1)+CD2*s^(n-2)
+CDn*s^(0))\n" + ...
    "        Where CN denotes Numerator constant value & CD denotes Denominator
constant value\n" + ...
    "        Input values for CN and CD in the form [C1 C2 C3... Cn]\n\n" + ...
    "\n
*****
*\n\n");
syms s;
prompt=["How many transfer functions do you want to evlauate: "];
N_Tf_2_Eval=input(prompt(1));
for i=1:N_Tf_2_Eval
prompt=["Is transfer Function in polynomial of Y/N: ", "Input single nummerical value for
K: ", "Input zero values Z: ", "Input pole values: "];
Poly_answer=input(prompt(1), "s");
K_val=input(prompt(2));
if Poly_answer == "Y"
    prompt=["Input Numerator constant values: ", "Input Denominator constant values: "];
    Num_val=input(prompt(1));
    Den_val=input(prompt(2));
    figure(i)
    G_s(i)=K_val*tf(Num_val,Den_val)
    Gs=zpk(G_s(i))
    margin(G_s(i)),grid
    [Gm,Pm,Wcg,Wcp]=margin(G_s(i))

else
    Z_val=input(prompt(3));
    P_val=input(prompt(4));

```



```
figure(i)
G_s(i)=tf(zpk(Z_val,P_val,K_val))
Gs=zpk(G_s(i))
margin(G_s(i)),grid,
[Gm,Pm,Wcg,Wcp]=margin(G_s(i))
end
end
```

